

Data Profiling & Analysis

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Step 1: Loading the Dataset

Dataset: tableau_prep_cleaning_exercise.csv | 100 rows × 18 columns

Step 2: Profile the Dataset (Before Cleaning)

2a. Missing Values

Column	Null Count
Satisfaction_Score	15
Payment_Method	14
Customer_Segment	12
Email	10
Notes	10
Age	7
Order_Date	5
City	3

2b. Data Type Issues

Column	Tableau Prep Infers	Correct Type
Order_Date	String	Date
Ship_Date	String	Date
All others	Correct	Correct

2c. Value Distributions

Column	Issue
Country	Language variants (Deutschland/Germany, España/Spain) + case
Product_Category	Case noise: 'electronics', 'ELECTRONICS', 'Electronics'
Customer_Segment	Case noise: 'CORPORATE', 'corporate', 'Corporate'
Payment_Method	Case noise: 'bank transfer', 'Bank Transfer'
Customer_Name	Same person in different cases/spacing
Unit_Price	Sentinel value 9999.99 in 5 rows
Age	2 rows with Age = 130 (impossible)
Quantity	4 rows with Quantity = 0

2d. Duplicates

- 5 exact duplicate rows found (rows 96–100 are copies of rows 5, 18, 29, 49, 65)
- All 5 share the same Order_ID as their originals — confirmed duplicates, not re-orders

2e. Cardinality of Key Columns

Column	Unique Count
Order_ID	95 of 100
Customer_ID	46
Customer_Name	39

Email	15
Country	15
Product_Category	10
Unit_Price	93

Step 3: Column Analysis

3a. Which columns have many unique values?

Column	Unique
Sales	92
Unit_Price	93
Order_Date	82
Ship_Date	87
Customer_ID	46
Age	43

3b. Which columns may be used as identifiers?

Column	Row Identifier?
Order_ID	YES (after dedup)
Customer_ID	No
Email	YES
Customer_Name	No

3c. Validity of Dates, Emails, Categories, and Numerics

Field	Issue
Order_Date	5 date formats; 5 null rows
Ship_Date	'not available' string
Email	10 nulls, 0 invalid patterns
Country	Language + case variants
Product_Category	Case variants only
Customer_Segment	Case variants + nulls
Quantity = 0	Zero-quantity orders
Age = 130	Impossible age value
Unit_Price = 9999.99	Sentinel/placeholder

Step 4: Structure Analysis

4a. Can Order_ID identify records?

- These 5 duplicates are exact full-row copies, same Order_ID, same values, same everything
- After deduplication: 95 rows, 95 unique Order_IDs to Order_ID is a valid primary key

4b. Can Customer_ID identify records?

- 46 unique Customer_IDs across 100 rows , customers appear multiple times
- Customer_ID is a foreign key (many-to-one), not a row identifier

4c. Does the Sales formula hold?

Formula to verify: $\text{Sales} = \text{Quantity} \times \text{Unit_Price} \times (1 - \text{Discount})$

- For rows with real Unit_Price values: the formula is consistent. So, Sales is correctly derived
- For 5 rows where Unit_Price = 9999.99 (sentinel): Sales is inflated (up to €56,999.94)

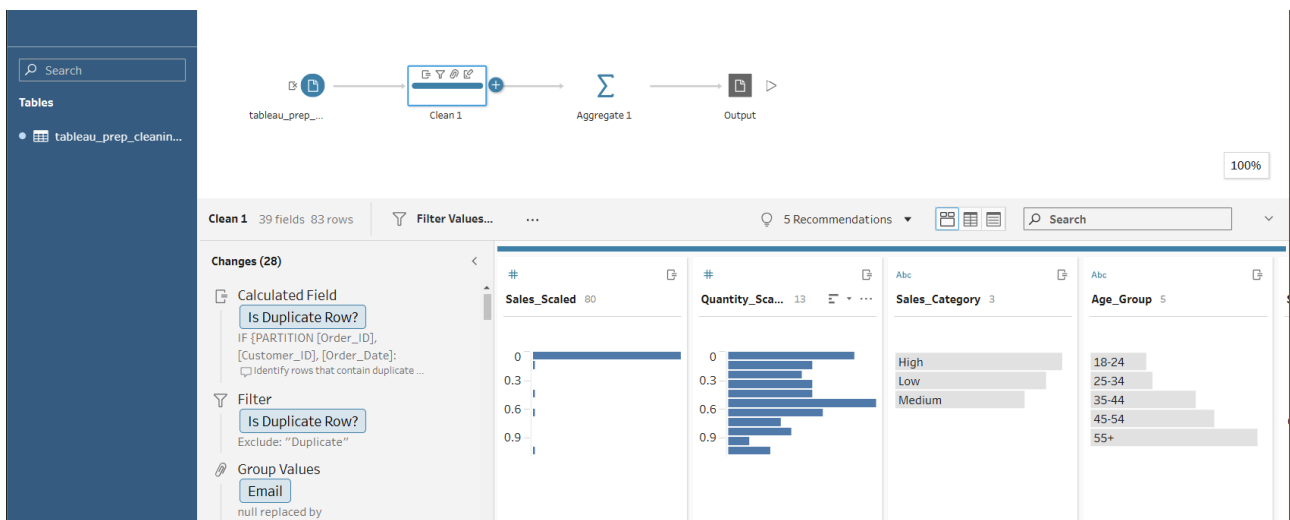
4d. Is Customer_Name consistent per Email?

- 13 emails map to more than one Customer_Name variant
- Example: mehdi.karimi@example.com is 'MEHDI KARIMI', 'Mehdi Karimi', 'mehdi karimi', 'Mehdi Karimi ' (trailing space)

Step 5: Cleaning Rules

#	Issue	Column(s)	Action	Logic
1	Duplicates	All	Remove	Exact matches (Order ID) are redundant.
2	Missing Dates	Order_Date	Drop Row	Can't analyze a sale without a date.
3	Mixed Formats	Dates	DATEPARSE	Standardize weird strings into Date types.
4	"Not Available"	Ship_Date	Set to Null	It's an optional field; better empty than text.
5	Country Typos	Country	Group/Map	Merged "Deutschland" and "Germany" into one.
6	Case/Spaces	Various	Trim & Proper	Fixed "city ", "CITY", and "City" into one format.
7	Null Segment	Cust_Segment	Mode (Corp)	Assigned the most common value (Corporate).
8	Null Payment	Payment	Mode (Bank)	Assigned "Bank Transfer" to fill gaps.
9	Age = 130	Age	Median (45)	Overwrote impossible ages with the middle.
10	Null Age	Age	Median (45)	Standard fill for missing demographics.
11	Price = 9999	Price	Cat. Median	Replaced "sentinel" values with category average.
12	Null Price	Price	Cat. Median	Grouped by category to get a realistic price.
13	Sales Errors	Sales	Recalculate	Redid math: Qty * Price * (1-Disc).
14	Qty = 0	Quantity	Drop Row	A zero-item order is an error.
15	Null Notes	Notes	"Unknown"	Categorical fill so it shows up in filters.
16	Null Score	Sat_Score	Leave Null	Don't fake customer feedback.
17	Null City/Email	City, Email	Leave Null	Not essential for the dashboard.

Step 6: Tableau Transformations



Step 7 & 8: Validation and Export

- Row Check: Confirmed row count dropped (due to duplicates/null dates) and matches our expected "clean" count.
 - Data Types: Verified in Prep that Dates are Calendar icons and Numbers are Hash icons.
 - Final Output: Exported as a .hyper file for Tableau Desktop to keep the dashboard fast.
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Step 9 : Pipeline Context: ETL vs. ELT

Which approach is better for this dataset?

ETL (Tableau Prep approach) is better if:

1. The dataset is moderately sized (millions of rows, not billions).
2. Data privacy is a concern (you want to mask or remove PII *before* it lands in storage).
3. The users are business analysts who prefer a visual interface over writing complex SQL inside a data warehouse.